

Research Gaps in Agentic LLM DI-AKI Prediction

An agentic LLM–time series framework for predicting drug-induced acute kidney injury using privacy-preserving synthetic EHR data is scientifically plausible, but the literature shows clear gaps in etiology-specific labeling, synthetic longitudinal validation, secure multi-institutional deployment, and prospective clinical testing. Existing work is much stronger on general AKI prediction than on drug-induced AKI, much stronger on structured time-series models than on agentic LLM systems, and much stronger on synthetic-data generation than on proving that synthetic data supports safe, calibrated, drug-specific AKI prediction in practice (Tomašev et al., 2019; Yousif & Awdishu, 2022; Giuffré & Shung, 2023; Liu et al., 2024).

General AKI models already achieve strong early prediction from longitudinal EHRs, including recurrent or continual models, multicenter time-series systems, and real-time hospital-wide models (Tomašev et al., 2019; Kate et al., 2020; Heo et al., 2023; Zhang et al., 2025). Drug-focused AKI prediction is much thinner: a few studies target nephrotoxic exposures such as cisplatin or vancomycin, and reviews explicitly note that most AKI models do not ascertain AKI etiology, making drug attribution difficult (Ambe et al., 2025; Aghamirzaei et al., 2025; Yousif & Awdishu, 2022).

1. Introduction

The target framework sits at the intersection of five partially connected literatures: dynamic AKI prediction, drug-induced AKI phenotyping, LLM-based clinical reasoning, synthetic EHR generation, and privacy-preserving distributed learning. Each subfield has advanced, but their combination remains largely unvalidated end-to-end (Shi et al., 2025; Shi et al., 2024; Liu et al., 2024; Pati et al., 2024).

The strongest foundation is dynamic EHR-based AKI prediction. Large recurrent and continual models can predict AKI hours to days before clinical recognition, and multicenter time-series models can retain good discrimination while providing some interpretability (Tomašev et al., 2019; Kate et al., 2020; Heo et al., 2023). But systematic reviews still find high risk of bias, sparse external validation, weak calibration reporting, and limited reproducibility across AKI prediction studies (Vagliano et al., 2022; Shi et al., 2025).

Are major research gaps still present in agentic LLM–time series frameworks for privacy-preserving prediction of drug-induced acute kidney injury?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on unresolved gaps in the target framework

The consensus is yes: the core pieces exist, but they have rarely been integrated and almost never validated together for drug-induced AKI. The missing evidence is concentrated in causal labeling, synthetic longitudinal fidelity, privacy-risk measurement, and deployment-ready evaluation (Yousif & Awdishu, 2022; Liu et al., 2024; Das et al., 2026).

2. Methods

This deep search ran over more than 170 million research papers indexed in Consensus across Semantic Scholar, PubMed, and other sources. The workflow identified 1,529 papers in the initial search, expanded through targeted sub-searches and citation crawling, screened 239, assessed 168 as eligible after relevance filtering, and included the top 50 most relevant papers for synthesis.

Search Strategy

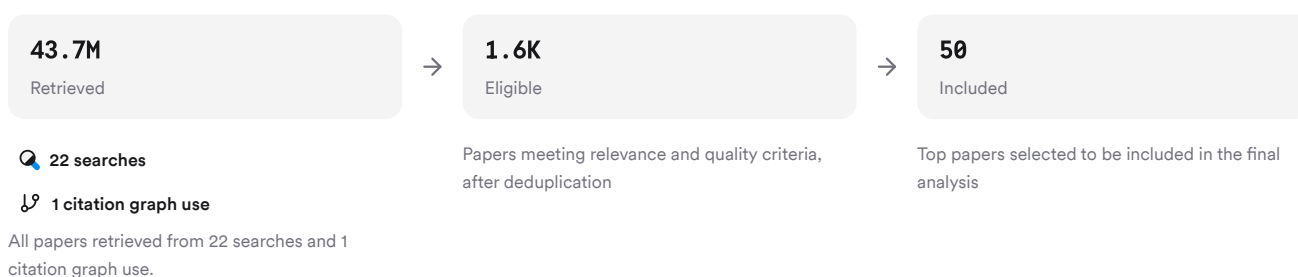


FIGURE 2 Deep search screening and inclusion workflow

Searches combined AKI prediction, drug-induced nephrotoxicity, LLM agents, time-series EHR modeling, synthetic data, and privacy-preserving learning.

3. Results

3.1 Key Papers

A few papers anchor the evidence because they define the state of the art in the adjacent pieces the target framework would need to combine. Large-scale dynamic AKI prediction is anchored by Tomašev et al., multicenter drug-specific time-series prediction by Heo et al., and etiology-specific DI-AKI design requirements by Yousif and Awdishu (Tomašev et al., 2019; Heo et al., 2023; Yousif & Awdishu, 2022).

Paper	Summary
(Kate et al., 2020)	Large-scale continuous AKI prediction foundation (Tomašev et al., 2019)
(Tomašev et al., 2019)	Multicenter drug-linked time-series AKI modeling (Heo et al., 2023)
(Zhao et al., 2025)	Defines DI-AKI phenotyping requirements (Yousif & Awdishu, 2022)

FIGURE 3 Foundational papers for the target framework

3.2 Etiology and Labeling

The biggest domain gap is that most AKI models predict AKI generally, not drug-induced AKI specifically (Yousif & Awdishu, 2022). Reviews of DI-AKI stress that expert adjudication is needed because temporal association alone misclassifies causality, DI-AKI varies by drug and mechanism, and predictors should include dose, concentrations, and interaction terms (Yousif & Awdishu, 2022).

Drug-specific studies remain fragmented by exposure. Cisplatin, vancomycin, contrast, and selected nephrotoxic combinations have each been modeled separately, but cross-drug representations and transferable etiology-aware phenotypes are still sparse (Ambe et al., 2025; Aghamirzaei et al., 2025; Li et al., 2022; Akimoto et al., 2023).

3.3 Synthetic Data and Privacy

Synthetic data studies show that high utility is achievable, sometimes with near-real downstream performance and low measured privacy risk (Yoon et al., 2023; Li et al., 2021; Venugopal et al., 2022). But reviews consistently say the hard cases remain longitudinal, high-dimensional, and bias-sensitive data, exactly the setting needed for medication-linked AKI trajectories (Liu et al., 2024; Giuffré & Shung, 2023).

Synthetic-data evaluation also lacks consensus standards. Multiple reviews call for transparent, standardized utility and privacy benchmarks, stronger regulatory alignment, and clearer reporting of re-identification risk and chain-of-custody safeguards (Arora et al., 2024; Giuffré & Shung, 2023; Susser et al., 2024).

Drug-AKI examples using synthetic augmentation are promising but preliminary. One small 2025 study found synthetic augmentation improved 30-day AKI prediction in vancomycin-ceftazidime/avibactam patients, but it also emphasized that further validation is warranted (Ramazani et al., 2025).

3.4 Agents, Multimodality, and Deployment

LLM and multi-agent systems can improve interpretability and sometimes external performance, but the evidence is early and mostly task-specific. AKI-Detector combined ML, LLMs, and RAG for ICU AKI prediction, while other LLM studies reported human-readable explanations, stronger external generalization, or useful text embeddings (Shi et al., 2024; Zhu et al., 2025; Sharabiani et al., 2025).

The remaining gaps are reliability and workflow integration. LLM studies explicitly note hallucination risk, unresolved probability calibration, and the need to test RAG or additional agents rather than assuming safety (Zhu et al., 2025). Broader reviews of agentic or human-AI co-design frameworks also highlight the importance of human oversight for leakage detection, bias detection, and concept refinement (Feng et al., 2026).

Results Timeline

AKI prediction research has moved from static risk models to dynamic, multimodal, and privacy-aware systems.

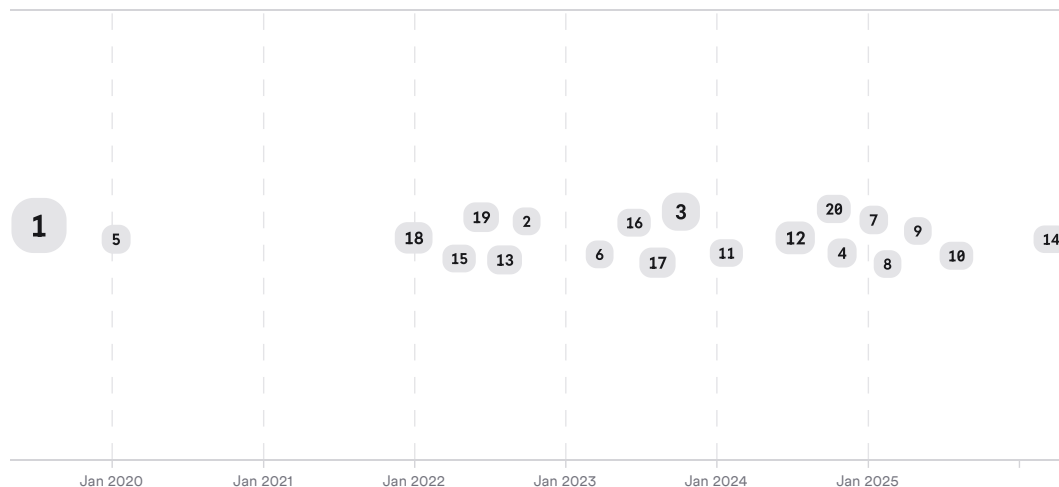


FIGURE 4 Timeline of included studies, larger markers indicate more citations

The arc is clear: early work established dynamic EHR prediction, later studies added multicenter time series and drug features, and the newest work layers LLMs, synthetic data, and federated privacy mechanisms on top. What is still missing is a mature study that combines all of these components for DI-AKI with robust external and prospective validation (Koyner et al., 2018; Heo et al., 2023; Manalu et al., 2024; Zhao et al., 2025).

Top Contributors

Type	Name	Papers
Author	Tongyue Shi	[0d787139e9b85097b4fb908ff9c1abe2][5aa61637d6d25119874b40f4dc2aeba7]
Author	Huiying Zhao	[0d787139e9b85097b4fb908ff9c1abe2][5aa61637d6d25119874b40f4dc2aeba7]
Author	Guilan Kong	[0d787139e9b85097b4fb908ff9c1abe2][5aa61637d6d25119874b40f4dc2aeba7]
Journal	<i>NPJ Digital Medicine</i>	[9499e0a992115b3a9c7982a4f4084453][96ff262c5b1c5b0390dee2aea503404e] [b54ac4b54be659b8b9780cdf2dd5935f]
Journal	<i>Nature</i>	[5796313001105d4fa2b06b1addab8bca]
Journal	<i>Journal of Biomedical Informatics</i>	[2ad8cae00db85385b47ccb0dc134fef0][4f613a9c85645e1ca7d52c5a4989afa1]

FIGURE 5 Authors and journals appearing most in this corpus

4. Discussion

The evidence base strongly supports dynamic EHR modeling for AKI, and moderately supports adding drug exposure features, multimodal notes, or interpretability layers. Large datasets, multicenter validation, and real-time framing are now common enough to show technical feasibility (Tomašev et al., 2019; Dong et al., 2021; Zhang et al., 2025; Manalu et al., 2024).

The evidence is much weaker for the full target framework. Drug-induced AKI requires better causal phenotyping than general AKI models typically provide, and most published LLM or agentic studies address ICU AKI or postoperative AKI rather than medication-specific nephrotoxicity (Yousif & Awdishu, 2022; Shi et al., 2024; Zhu et al., 2025). Synthetic EHR work shows promise, but reviews still point to unresolved temporal fidelity, subgroup bias, privacy benchmarking, and regulatory accountability (Liu et al., 2024; Arora et al., 2024).

Privacy-preserving deployment is also unresolved. Federated learning can improve collaboration without raw-data sharing, but exchanged updates can still leak information, and current FL research still has gaps in fairness, personalization, and communication efficiency (Zhao et al., 2025; Pati et al., 2024; Das et al., 2026).

Claim	Evidence Strength	Reasoning	Papers
Dynamic time-series EHR models can predict AKI early and accurately.	 Strong	Replicated across large, multicenter, and real-time studies.	(Tomašev et al., 2019; Zhang et al., 2025; Xu et al., 2022)
Drug-specific AKI prediction is underdeveloped relative to general AKI prediction.	 Moderate	Multiple reviews and disease-specific papers identify this gap.	(Ambe et al., 2025; Yousif & Awdishu, 2022; Manalu et al., 2024)
Synthetic EHR data can preserve utility, but longitudinal bias and privacy evaluation remain unresolved.	 Moderate	Several strong synthetic-data papers support utility, while reviews stress remaining limitations.	(Yoon et al., 2023; Li et al., 2021; Liu et al., 2024)
Agentic LLM systems appear promising for explanation and reasoning in AKI prediction.	 Moderate	Early studies report gains, but evidence is recent and sparse.	(Shi et al., 2024; Zhu et al., 2025; Li et al., 2025)
End-to-end privacy-preserving agentic DI-AKI frameworks are clinically validated.	 Weak	The component technologies exist, but direct integrated validation is largely absent.	(Qian et al., 2023; Das et al., 2026; Yousif & Awdishu, 2022)

FIGURE 6 Key claims and supporting evidence in the corpus

5. Conclusion

The literature supports the ingredients of the proposed framework, but not the finished system. Research has advanced furthest on dynamic AKI prediction and furthest least on proving that agentic LLMs plus synthetic EHRs can deliver trustworthy, privacy-preserving, drug-specific AKI prediction across institutions.

Research Gaps

The most important gaps are concentrated in causality, data realism, deployment security, and clinical usefulness.

Topic/Outcome	Etiology adjudication	Longitudinal synthetic fidelity	External prospective validation	Privacy attack testing
General AKI prediction	14	4	6	2
Drug-induced AKI prediction	3	2	2	1
LLM or agentic AKI systems	2	GAP	1	GAP

Topic/Outcome	Etiology adjudication	Longitudinal synthetic fidelity	External prospective validation	Privacy attack testing
Synthetic EHR for clinical prediction	1	8	3	5
Federated privacy-preserving EHR prediction	1	1	4	7

The sharpest gap is the near absence of studies that jointly evaluate DI-AKI causality, agentic reasoning, synthetic longitudinal data, and formal privacy threats in one framework. That gap persists because each piece has its own validation burden, and current studies usually optimize one axis at a time rather than the full socio-technical system (Yousif & Awdishu, 2022; Liu et al., 2024; Pati et al., 2024; Susser et al., 2024).

Open Research Questions

The next useful studies would test whether the combined framework is not just accurate, but causally credible, privacy-robust, and clinically deployable.

Question	Why
Can expert-adjudicated, drug-specific AKI labels improve agentic time-series model validity more than larger but weakly labeled cohorts?	DI-AKI reviews argue misclassification of etiology is a central limitation and likely constrains all downstream modeling.
Do synthetic longitudinal EHRs preserve medication timing, dose-response, and interaction signals well enough for DI-AKI prediction?	Reviews identify temporal correlation and bias preservation as major unresolved issues for synthetic healthcare data.
Does combining synthetic data with federated learning and RAG-based agents improve privacy and generalization without harming calibration?	The field lacks integrated comparisons across privacy utility, hallucination control, and real-world transportability.

In short, the main research gaps in this agentic LLM–time series DI-AKI framework are not whether the components work individually, but whether they work together safely, causally, and reproducibly.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

- Aghamirzaei, F., Abin, A. A., & Futuhi, F. (2025). An Ensemble Machine Learning Model for Early Prediction of Vancomycin-Induced Acute Kidney Injury in ICU Patients. *Archives of Academic Emergency Medicine*, 13. <https://doi.org/10.22037/aaemj.v13i1.2560>
- Akimoto, H., Hayakawa, T., Nagashima, T., Minagawa, K., Takahashi, Y., & Asai, S. (2023). Detection of potential drug-drug interactions for risk of acute kidney injury: a population-based case-control study using interpretable machine-learning models. *Frontiers in Pharmacology*, 14. <https://doi.org/10.3389/fphar.2023.1176096>
- Ambe, K., Aoki, Y., Murashima, M., Wachino, C., Deki, Y., Ieda, M., Kondo, M., Furukawa-Hibi, Y., Kimura, K., Hamano, T., & Tohkin, M. (2025). Prediction of Cisplatin-Induced Acute Kidney Injury Using an Interpretable Machine Learning Model and Electronic Medical Record Information. *Clinical and Translational Science*, 18. <https://doi.org/10.1111/cts.70115>
- Arora, A., Wagner, S. K., Carpenter, R., Jena, R., & Keane, P. (2024). The urgent need to accelerate synthetic data privacy frameworks for medical research.. *The Lancet. Digital health*. [https://doi.org/10.1016/s2589-7500\(24\)00196-1](https://doi.org/10.1016/s2589-7500(24)00196-1)
- Das, B., Dutta, P., & Mandal, A. (2026). Secure and Privacy-Preserving Federated Learning for Healthcare Time Series Data: A Comprehensive Review. *2026 Innovations in Machine, Engineering, and Digital Conference (IMED)*, 1-5. <https://doi.org/10.1109/imed68921.2026.11484225>

- Dong, J., Feng, T., Thapa-Chhetry, B., Cho, B. G., Shum, T., Inwald, D., Newth, C., & Vaidya, V. (2021). Machine learning model for early prediction of acute kidney injury (AKI) in pediatric critical care. *Critical Care*, 25. <https://doi.org/10.1186/s13054-021-03724-0>
- Feng, J., Kothari, A., Vossler, P., Bishara, A. M., Zier, L., Addo, N., Kornblith, A., Tan, Y., & Singh, C. (2026). Human-AI Co-design for Clinical Prediction Models. *NPJ digital medicine*. <https://doi.org/10.1038/s41746-026-02838-5>
- Giuffré, M., & Shung, D. (2023). Harnessing the power of synthetic data in healthcare: innovation, application, and privacy. *NPJ Digital Medicine*, 6. <https://doi.org/10.1038/s41746-023-00927-3>
- Heo, S., Kang, E., Yu, J., Kim, H., Lee, S., Kim, K., Hwangbo, Y., Park, R. W., Shin, H., Ryu, K., Kim, C., Jung, H., Chegal, Y., Lee, J.-H., & Park, Y. (2023). Time Series AI Model for Acute Kidney Injury Detection Based on a Multicenter Distributed Research Network: Development and Verification Study. *JMIR Medical Informatics*, 12. <https://doi.org/10.2196/47693>
- Kate, R. J., Pearce, N., Mazumdar, D., & Nilakantan, V. (2020). A continual prediction model for inpatient acute kidney injury. *Computers in biology and medicine*, 116, 103580. <https://doi.org/10.1016/j.compbiomed.2019.103580>
- Koyner, J., Carey, K., Edelson, D., & Churpek, M. (2018). The Development of a Machine Learning Inpatient Acute Kidney Injury Prediction Model*. *Critical Care Medicine*, 46, 1070–1077. <https://doi.org/10.1097/ccm.0000000000003123>
- Li, Y., Chan, T.-M., Feng, J., Tao, L., Jiang, J., Zheng, B., Huo, Y., & Li, J. (2022). A pattern-discovery-based outcome predictive tool integrated with clinical data repository: design and a case study on contrast related acute kidney injury. *BMC Medical Informatics and Decision Making*, 22. <https://doi.org/10.1186/s12911-022-01841-6>
- Li, J., Cairns, B., Li, J., & Zhu, T. (2021). Generating synthetic mixed-type longitudinal electronic health records for artificial intelligent applications. *NPJ Digital Medicine*, 6. <https://doi.org/10.1038/s41746-023-00834-7>
- Li, Z., Wang, L., Zhang, X., Wu, A., & Liu, T. (2025). Integration of machine learning and large language models for screening and identifying key risk factors of acute kidney injury after cardiac surgery. *Frontiers in Medicine*, 12. <https://doi.org/10.3389/fmed.2025.1618222>
- Liu, Y., Acharya, U. R., & Tan, J. H. (2024). Preserving privacy in healthcare: A systematic review of deep learning approaches for synthetic data generation. *Computer methods and programs in biomedicine*, 260, 108571. <https://doi.org/10.1016/j.cmpb.2024.108571>
- Manalu, G. D. M., Christian, M. M., You, S., & Choi, H. (2024). Enhancing Acute Kidney Injury Prediction through Integration of Drug Features in Intensive Care Units. *ArXiv, abs/2401.04368*. <https://doi.org/10.7236/ijasc.2023.12.4.434>
- Pati, S., Kumar, S., Varma, A., Edwards, B., Lu, C., Qu, L., Wang, J., Lakshminarayanan, A., Wang, S.-H., Sheller, M. J., Chang, K., Singh, P., Rubin, D. L., Kalpathy-Cramer, J., & Bakas, S. (2024). Privacy preservation for federated learning in health care. *Patterns*, 5. <https://doi.org/10.1016/j.patter.2024.100974>
- Qian, Z., Callender, T., Cebere, B., Janes, S. M., Navani, N., & Van Der Schaar, M. (2023). Synthetic data for privacy-preserving clinical risk prediction. *Scientific Reports*, 14. <https://doi.org/10.1038/s41598-024-72894-y>
- Ramazani, M., Brothers, T., Ahmed, I., & Al-Mamun, M. A. (2025). Synthetic Data–Driven Early Prediction Framework for Acute Kidney Injury in Patients Receiving Vancomycin and Ceftazidime/Avibactam. *Pharmacotherapy: The Journal of Human Pharmacology and Drug Therapy*, 46. <https://doi.org/10.1002/phar.70064>
- Sharabiani, M., Mahani, A. S., Bottle, A., Srinivasan, Y., Issitt, R., & Stoica, S. (2025). GenAI exceeds clinical experts in predicting acute kidney injury following paediatric cardiopulmonary bypass. *Scientific Reports*, 15. <https://doi.org/10.1038/s41598-025-04651-8>
- Shi, T., Lin, Y., Zhao, H., & Kong, G. (2025). Artificial intelligence models for predicting acute kidney injury in the intensive care unit: a systematic review of modeling methods, data utilization, and clinical applicability. *JAMIA Open*, 8. <https://doi.org/10.1093/jamiaopen/ooaf065>
- Shi, T., Xiao, M., Xu, H., Zhao, H., & Kong, G. (2024). AKI-Detector: A Multi-Agent Framework by Integrating Machine Learning and Large Language Models for Early Prediction of Acute Kidney Injury in ICU.. *AMIA ... Annual Symposium proceedings. AMIA Symposium*, 2024, 1190-1199.
- Susser, D., Schiff, D., Gerke, S., Cabrera, L. Y., Cohen, I. G., Doerr, M., Harrod, J., Kostick-Quenet, K., McNealy, J. E., Meyer, M. N., Price, W., & Wagner, J. K. (2024). Synthetic Health Data: Real Ethical Promise and Peril. *The Hastings Center Report*, 54, 8 - 13. <https://doi.org/10.1002/hast.4911>

Tomašev, N., Glorot, X., Rae, J. W., Zielinski, M., Askham, H., Saraiva, A., Mottram, A., Meyer, C., Ravuri, S. V., Protsyuk, I. V., Connell, A., Hughes, C. O., Karthikesalingam, A., Cornebise, J., Montgomery, H., Rees, G., Laing, C., Baker, C., Peterson, K. S., . . . Mohamed, S. (2019). A Clinically Applicable Approach to Continuous Prediction of Future Acute Kidney Injury. *Nature*, 572, 116 - 119.

<https://doi.org/10.1038/s41586-019-1390-1>

Vagliano, I., Chesnaye, N., Leopold, J., Jager, K., Abu-Hanna, A., & Schut, M. (2022). Machine learning models for predicting acute kidney injury: a systematic review and critical appraisal. *Clinical Kidney Journal*, 15, 2266 - 2280. <https://doi.org/10.1093/ckj/sfac181>

Venugopal, R., Shafqat, N., Venugopal, I., Tillbury, B. M. J., Stafford, H. D., & Bourazeri, A. (2022). Privacy preserving Generative Adversarial Networks to model Electronic Health Records. *Neural networks : the official journal of the International Neural Network Society*, 153, 339-348. <https://doi.org/10.1016/j.neunet.2022.06.022>

Xu, J., Hu, Y., Liu, H., Mi, W., Li, G., Guo, J., & Feng, Y. (2022). A novel multivariable time series prediction model for acute kidney injury in general hospitalization. *International journal of medical informatics*, 161, 104729.

<https://doi.org/10.1016/j.ijmedinf.2022.104729>

Yoon, J., Mizrahi, M., Ghalaty, N. F., Jarvinen, T., Ravi, A. S., Brune, P., Kong, F., Anderson, D., Lee, G., Meir, A., Bandukwala, F., Kanal, E., Arik, S. Ö., & Pfister, T. (2023). EHR-Safe: generating high-fidelity and privacy-preserving synthetic electronic health records. *NPJ Digital Medicine*, 6. <https://doi.org/10.1038/s41746-023-00888-7>

Yousif, Z. K., & Awdishu, L. (2022). Drug-Induced Acute Kidney Injury Risk Prediction Models. *Nephron*, 147, 44 - 47.

<https://doi.org/10.1159/000526267>

Zhang, Y., Xu, D., Gao, J., Wang, R., Yan, K., Liang, H., Xu, J., Zhao, Y., Zheng, X., Xu, L., Wang, J., Zhou, F., Zhou, G., Zhou, Q., Yang, Z., Chen, X., Shen, Y., Ji, T., Feng, Y., . . . Yang, L. (2025). Development and validation of a real-time prediction model for acute kidney injury in hospitalized patients. *Nature Communications*, 16. <https://doi.org/10.1038/s41467-024-55629-5>

Zhao, H., Sui, D., Wang, Y., L., & Wang, L. (2025). Privacy-Preserving Federated Learning Framework for Multi-Source Electronic Health Records Prognosis Prediction. *Sensors (Basel, Switzerland)*, 25. <https://doi.org/10.3390/s25082374>

Zhu, H., Wang, R., Qian, J., Wu, Y., Jin, Z., Shan, X., Ji, F., Yuan, Z., & Pan, T. (2025). Leveraging Large Language Models for Predicting Postoperative Acute Kidney Injury in Elderly Patients. *BME Frontiers*, 6. <https://doi.org/10.34133/bmef.0111>